



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : BIO G532
Course Title : BIOSTATISTICS & BIOMODELING
Instructor-in-Charge : SUSHMITA BASU
Instructor : Shibasish Chowdhury

1. Course Description: Biostatistics and Biomodeling course has two components. Biostatistics part will provide an introduction to the basic concepts of statistical methods to equip students with statistical reasoning in their fields of study. Topics include variables in biology, Standard deviation; correlation and regression analysis; significance tests; chi square, t-test, ANOVA etc. Emphasis is placed on the application of fundamental concepts to real world situations. Biomodeling is designed to provide students the first-hand experience of modelling biomolecules to elucidate the relationships between sequence, structure and function. Specifically, different computing strategies for the simulation of biomolecules will be discussed with regard to their utility, advantages, and limitations.

2. Scope and Objective of the Course: The student would learn how to collect data suitable for statistical analyses, understand the general features of the data collected, making statistical inferences and likelihood of occurrence of an event using probability calculations and tests of significance.

3. Text Books:

- Biostatistics: A foundation for Analysis in the Health Sciences. Wayne Daniel, Ninth Edition, Wiley-India Edition, 2009.
- Molecular modeling: Principles and Applications. Andrew R Leach, 2nd Edition, 2001, Pearson Education Lim.

4. Reference Books:

- Principles of Biostatistics, M. Pagano and K. Gauvereau, Thomson Learning Publ., Indian Ed., 2007.
- Molecular Modeling and Simulation - An Interdisciplinary Guide. Tamar Schlick, Springer, New York, 2002.

5. Course Plan:

Module / Lecture	Lecture Session	Text/Reference Book (Sec/Chap)	Learning outcomes
M0-L1	Introduction	-	Scope and application of statistics and modeling in modern biology
M1.1-L1-5	Protein structure and conformation	R-2 (3-4)	Protein structural hierarchy, Structural motifs, classification



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

M1.1-L6	Molecular graphics	Class Notes and Database resources	Introduction to graphic representation, Representation of molecular structure: macro-molecules Database of macro-molecular structures
M1.1- L7-9	Protein structure prediction	T-2(10)	First principle methods for predicting protein structure, comparative modeling, threading, CASP
M1.2- L10-13	Conformation and Variability in DNA structures	R-2 (5-6)	Conformational Flexibility, Canonical DNA Forms, DNA Sequence Effects, DNA local geometry, base-pair parameters, base step parameters, DNA stiffness
M1.3- L14-17	Force-field and MD simulation	T-2(4), T-2(6-7), class notes	Energy minimization and MD simulation techniques
M2.1- L1-4	Descriptive statistics: Understanding the general features of the data collected for a study	T1 (2)	Types of data, Scale Measurements; Representation of variables; Measures of Central Tendency, Measures of Dispersion, other averages, Box-plot
M2.1- L5-8	Probability Distribution	T1 (3,4)	Classical definition of Probability, addition and multiplication rules. Random Variables & Discrete Counts (Binomial Distribution), Continuous Variables & The Normal Distribution; their means and variances.
M2.2/ L9-13	Inferential statistics: foundations and parametric tests	T1 (6,7,8)	Foundations of hypothesis testing: Notion of hypothesis, null and alternative, test statistic, critical region, idea of one and two tailed test, Type-I/II errors, p-value, t-test (independent and paired), ANOVA (One-way, two-way), Tukey's post hoc HSD.
M2.2- L14-17	Inferential statistics: non-parametric tests and categorical data	T1 (12,13)	Chi-squared test for categorical data, Mann-Whitney U test, Kruskal Wallis test.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

M2.3 - L18-20	Simple Linear Regression and Correlation	T1 (9)	Simple linear regression model: lines of regression, regression coefficient, Coefficient of determination. Scatter diagram, Karl Pearson's coefficient of correlation.
M2.4 – L21-23	Applied biostatistics – case study	Class notes	Application of statistical methods in real-world clinical/translational research.

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close / Open Book)
Mid-Semester Test	90 Min.	25		Closed Book
Comprehensive Examination	3 h	35		Closed Book/ Open Book
Practice based Laboratory Component/ Quizzes/ Assignments	Variable	40	Distributed through- out the semester	Closed/ Open Book

7. Chamber Consultation Hour: To be Announced

8. Notices: Notices pertaining to this course will be displayed on Bio Notice Board or Nalanda.

9. Make-up Policy: Make-up will be given only in case of hospitalization cases and under emergency conditions for which all documents need to be provided. It is expected that students shall avoid misuse of this feature. The decision of IC/ Course Instructor will be final.

10. Grading Procedure: Award of grades would be guided by the histogram of marks and course average. To receive a valid grade, a student must score **at least 40% of the median score or 30% of the average of the top 3 scores**, whichever is lower. **Failure to meet this requirement will result in an “NC” report for the course.**

Instructor-in-charge
Course No. BIO G532